Experiment 4

Experiment to understand data preprocessing in Data Science.

Code 1

```
import numpy as np
import pandas as pd

df = pd.read_csv("pre-process_datasample.csv")
print("Original Dataset:")
print(df)
print()

print("Dataset Info:")
df.info()
print()

print("Mode of Country column:")
print(df.Country.mode())
print()




df['Country'] = df.Country.fillna(df.Country.mode()[0])
df['Age'] = df.Age.fillna(df.Age.median())
df['Salary'] = df.Salary.fillna(round(df.Salary.mean()))
print("Dataset after handling missing values:")
print(df)
print()

print("One-Hot Encoded Country column:")
print(pd.get_dummies(df.Country))
print()

updated_dataset = pd.concat([pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
axis=1)
print("Updated dataset after One-Hot Encoding:")
print(updated_dataset)
print()
print("Updated Dataset Info:")
updated_dataset.info()
print()

updated_dataset['Purchased'] = updated_dataset.Purchased.replace(['No', 'Yes'], [0,
1])
print("Final Preprocessed Dataset:")
print(updated_dataset)
```

Output

```
Original Dataset:
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0      NaN       Yes
5   France  35.0  58000.0       Yes
6    Spain   NaN  52000.0        No
7   France  48.0  79000.0       Yes
8  Germany  50.0  83000.0        No
9   France  37.0  67000.0       Yes

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes

Mode of Country column:
0    France
```

```
Name: Country, dtype: object

Dataset after handling missing values:
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0  63778.0       Yes
5   France  35.0  58000.0       Yes
6    Spain  38.0  52000.0        No
7   France  48.0  79000.0       Yes
8  Germany  50.0  83000.0        No
9   France  37.0  67000.0       Yes

One-Hot Encoded Country column:
   France  Germany  Spain
0    True    False  False
1   False    False   True
2   False     True  False
3   False    False   True
4   False     True  False
5    True    False  False
6   False    False   True
7    True    False  False
8   False     True  False
9    True    False  False

Updated dataset after One-Hot Encoding:
   France  Germany  Spain   Age   Salary Purchased
0    True    False  False  44.0  72000.0        No
1   False    False   True  27.0  48000.0       Yes
2   False     True  False  30.0  54000.0        No
3   False    False   True  38.0  61000.0        No
4   False     True  False  40.0  63778.0       Yes
5    True    False  False  35.0  58000.0       Yes
6   False    False   True  38.0  52000.0        No
7    True    False  False  48.0  79000.0       Yes
8   False     True  False  50.0  83000.0        No
9    True    False  False  37.0  67000.0       Yes

Updated Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   France     10 non-null     bool
 1   Germany    10 non-null     bool
 2   Spain      10 non-null     bool
 3   Age        10 non-null     float64
 4   Salary     10 non-null     float64
 5   Purchased  10 non-null     object
dtypes: bool(3), float64(2), object(1)
memory usage: 402.0+ bytes

Final Preprocessed Dataset:
   France  Germany  Spain   Age   Salary  Purchased
0    True    False  False  44.0  72000.0          0
1   False    False   True  27.0  48000.0          1
2   False     True  False  30.0  54000.0          0
3   False    False   True  38.0  61000.0          0
4   False     True  False  40.0  63778.0          1
5    True    False  False  35.0  58000.0          1
6   False    False   True  38.0  52000.0          0
7    True    False  False  48.0  79000.0          1
8   False     True  False  50.0  83000.0          0
9    True    False  False  37.0  67000.0          1
```

Output

```
C:\Users\vsros\AppData\Local\Temp\ipykernel_20308\1860359933.py:40: FutureWarning: Downcasting behavior in `replace` is
  updated_dataset['Purchased'] = updated_dataset.Purchased.replace(['No', 'Yes'], [0,
```